Gyires's permanent inequality for nonnegative circulant matrices of order five

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Abstract

B. Gyires conjectured in 1978 that every doubly stochastic $n \times n$ matrix A satisfies $4\,\mathrm{per}(A)^2 \geq \frac{n!}{n^n}\big(\mathrm{per}(AA^{\mathsf{T}}) + \mathrm{per}(A^{\mathsf{T}}A) + 2\,\mathrm{per}(A^2)\big)$, with equality only at the matrix J_n all of whose entries are $1/n$; the conjecture is open for $n \geq 5$. We prove the conjecture for every doubly stochastic circulant matrix of order five, that is, for every convex combination of the five powers of the cyclic permutation matrix, including the matrices with zero entries. The proof establishes the stronger inequality $24\,\mathrm{per}(CC^{\mathsf{T}}) \leq 625\,\mathrm{per}(C)^2$ for every circulant C of order five with nonnegative entries, with equality only when all entries of C are equal, together with a quantitative form: if $z_1, \ldots, z_4$ are the gaps between consecutive values when the entries of the first row of C are sorted in increasing order, then $625\,\mathrm{per}(C)^2 - 24\,\mathrm{per}(CC^{\mathsf{T}}) \geq 484\,(z_1^{10} + z_2^{10} + z_3^{10} + z_4^{10})$, and the constant 484 is best possible. After an affine change of indices modulo five, the entries of the first row may be assumed to increase along one of six canonical orderings, and for each of them $625\,\mathrm{per}(C)^2 - 24\,\mathrm{per}(CC^{\mathsf{T}})$, written in the smallest entry and the gaps, is a polynomial with nonnegative integer coefficients. The results have been checked by computer and proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an AI research effort directed by Dev D. Goyal. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working autonomously from prompts by Dev D. Goyal, proposed the problem, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose which problems to pursue and asked for proofs, formal verification and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it ported the formal proofs to a current version of Lean, rebuilt them from source and confirmed that they rely only on Lean's three standard axioms, tested every claim with separately written exact computations, and read every step of the proofs. It also wrote the present version of this paper. Dev D. Goyal has reviewed the proofs and is responsible for them; this review is not an independent expert validation. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The conjecture

A real square matrix is *doubly stochastic* if its entries are nonnegative and all its row sums and column sums equal 1. We write Ω_n for the set of doubly stochastic $n \times n$ matrices, $J_n \in \Omega_n$ for the matrix all of whose entries equal $1/n$, and

$$\mathrm{per}\, A = \sum_{\sigma \in S_n} \prod_{i=1}^{n} a_{i\sigma(i)}$$

for the permanent of $A = (a_{ij})$. By van der Waerden's conjecture, proved independently by Gyires [6], Egorychev [4] and Falikman [5], $\operatorname{per} A \geq n!/n^n$ for every $A \in \Omega_n$, with equality only at J_n. B. Gyires [7] proposed a related lower bound for the square of the permanent. It is Conjecture 20 in the survey of open problems on permanents by Cheon and Wanless [3], in the following form, where A^* denotes the conjugate transpose.

Conjecture 1.1 (Gyires; Cheon and Wanless [3, Conjecture 20])**.** *For every $A \in \Omega_n$,*

$$\frac{4\operatorname{per}(A)^2}{\operatorname{per}(AA^*) + \operatorname{per}(A^*A) + 2\operatorname{per}(A^2)} \;\geq\; \frac{n!}{n^n},$$

with equality if and only if $A = J_n$.

For real matrices $A^* = A^{\mathsf{T}}$, the transpose, and this is the form used below. For $A \in \Omega_n$ the matrices AA^{T}, $A^{\mathsf{T}}A$ and A^2 are doubly stochastic, so by van der Waerden's theorem each of their permanents is at least $n!/n^n$, and the denominator is positive. The conjecture is therefore equivalent to the polynomial inequality

$$\frac{n!}{n^n}\bigl(\operatorname{per}(AA^{\mathsf{T}}) + \operatorname{per}(A^{\mathsf{T}}A) + 2\operatorname{per}(A^2)\bigr) \;\leq\; 4\operatorname{per}(A)^2 \qquad (A \in \Omega_n), \tag{1}$$

with equality only at J_n. We call (1) the *homogeneous form* of Gyires's inequality: both sides are homogeneous of degree $2n$ in the entries of A, so it makes sense for every square matrix.

1.2 What this paper proves

A matrix C of order five is *circulant* if its entries depend only on the difference of the indices modulo five: $c_{ij} = w_{j-i}$ for a vector $w = (w_0, \ldots, w_4) \in \mathbb{R}^5$, the indices being read in $\mathbb{Z}/5\mathbb{Z}$. We call w, the first row of C, its vector of *weights*, and write $C = C(w)$. A circulant is doubly stochastic exactly when $w \geq 0$ and $w_0 + \cdots + w_4 = 1$, so the doubly stochastic circulants of order five are the convex combinations $\sum_k w_k Q^k$ of the powers of the cyclic permutation matrix $Q = C(0,1,0,0,0)$. They form a four-dimensional simplex inside the sixteen-dimensional polytope Ω_5; this simplex contains matrices with zero entries, among them its vertices Q^k. For $n = 5$ the constant of the conjecture is $5!/5^5 = 24/625$.

Main results. *Let $C = C(w)$ be a circulant of order five with nonnegative weights.*

(a) (Theorem 3.1, the strong inequality.) $24\operatorname{per}(CC^{\mathsf{T}}) \leq 625\operatorname{per}(C)^2$, with equality if and only if all five weights are equal.

(b) (Theorem 3.2, the margin.) If $w_{\pi(0)} \leq w_{\pi(1)} \leq \cdots \leq w_{\pi(4)}$ for a permutation π and $z_j = w_{\pi(j)} - w_{\pi(j-1)}$ for $1 \leq j \leq 4$, then $625\operatorname{per}(C)^2 - 24\operatorname{per}(CC^{\mathsf{T}}) \geq 484\,(z_1^{10} + z_2^{10} + z_3^{10} + z_4^{10})$, and no constant larger than 484 is possible.

(c) (Corollary 3.3.) The homogeneous form (1) of Gyires's inequality holds for C, with equality if and only if all five weights are equal. In particular Conjecture 1.1 holds for every doubly stochastic circulant of order five, including its equality case.

The strong inequality (a) implies Gyires's inequality for C: for every real circulant C one has $CC^{\mathsf{T}} = C^{\mathsf{T}}C$ and $\operatorname{per}(C^2) \leq \operatorname{per}(CC^{\mathsf{T}})$ (Corollary 2.4), so the left-hand side of (1) is at most $\frac{24}{625} \cdot 4\operatorname{per}(CC^{\mathsf{T}})$, which by (a) is at most $4\operatorname{per}(C)^2$.

1.3 What was known

For $n = 2$ every matrix in Ω_2 has the form $\left(\begin{smallmatrix} a & 1-a \\ 1-a & a \end{smallmatrix}\right)$ with $0 \le a \le 1$. With $s = a^2 + (1-a)^2$ and $t = 2a(1-a)$ one finds $\operatorname{per}(A) = s$ and $\operatorname{per}(AA^{\mathsf{T}}) = \operatorname{per}(A^{\mathsf{T}}A) = \operatorname{per}(A^2) = s^2 + t^2$, so the conjecture reads $2s^2 \ge s^2 + t^2$, that is, $s \ge t$, or $(2a-1)^2 \ge 0$, with equality only at $a = \frac{1}{2}$, that is, at J_2. The inequality for $n = 3$ was proved by D. K. Chang [2]. In 2026 C. Ozols [11] proved the case $n = 4$ for all of Ω_4, including the equality case. That proof is computer-assisted and formalized in Lean; its finite certificate is checked by compiled code (`native_decide`) rather than by Lean's kernel alone. Ozols's formalization also proves the inequality $3\operatorname{per}(AA^{\mathsf{T}}) \le 32\operatorname{per}(A)^2$ on Ω_4, the analogue of (a) for $n = 4$, and the comparison of Lemma 2.3 below. Ozols's account of prior work [11] also reports that Cheon and Wanless [3] record a sufficient condition of Chang for every n: the inequality holds when every entry of A is at least $(n-2)/(n-1)^2$. For $n = 5$ this condition covers the doubly stochastic circulants all of whose weights lie in $[\frac{3}{16}, \frac{1}{4}]$, a neighbourhood of J_5, but no circulant with a zero weight. The same account reports that a catalogue of Minc's conjectures maintained by Y. Lavi lists the conjecture as open and announces results, stated without proofs, for matrices in Ω_4 with a zero entry, for matrices near J_n and for special families. That catalogue could not be consulted for this paper, and whether those special families include circulants is not known. No earlier treatment of the circulant case of order five, or of the inequalities (a) and (b), has been found; the searches for earlier work are described in the section "About this paper".

1.4 Motivation

Gyires's conjecture is open from order five on. Circulant matrices form a natural family on which to test it, and their permanents have long been studied (see, for instance, Minc [9] for doubly stochastic circulants of the form $\alpha I + \beta Q + \gamma Q^2$): the doubly stochastic circulants are exactly the doubly stochastic matrices invariant under the simultaneous cyclic shift of rows and columns, every real circulant C is normal, $CC^{\mathsf{T}} = C^{\mathsf{T}}C$, and the permanents in the conjecture become polynomials in five weights. The family is four-dimensional, so the result cannot be obtained by checking finitely many matrices, and it reaches the boundary of Ω_5, so it is not covered by results for matrices near J_5. The result shows that the conjecture holds on this family, with the margin (b), which vanishes only at the constant matrices, and the proof exhibits coordinates in which the relevant polynomial has nonnegative coefficients.

1.5 How to read this paper

Section 2 collects the identities for circulants and a comparison lemma for permanents. Section 3 states the results, and Section 4 proves them. Section 5 describes the computer checks and the formal proof, and Section 6 discusses the scope of the results.

2 Background

2.1 Notation

Indices of rows, columns and weights are taken in $\mathbb{Z}/5\mathbb{Z} = \{0, 1, 2, 3, 4\}$. For $w \in \mathbb{R}^5$ we write $C(w)$ for the circulant with entries $C(w)_{ij} = w_{j-i}$, and

$$P(w) = \operatorname{per} C(w), \qquad g(w)_k = \sum_{i=0}^{4} w_i w_{i+k}, \qquad h(w)_k = \sum_{i=0}^{4} w_i w_{k-i} \qquad (k \in \mathbb{Z}/5\mathbb{Z}),$$

so that $g(w)$ is the circular autocorrelation of w and $h(w)$ its circular self-convolution. Note that $g(w)_{-k} = g(w)_k$. The *defects*, D_s for the strong inequality and D_G for Gyires's inequality, are the homogeneous polynomials of degree ten

$$D_s(w) = 625\,P(w)^2 - 24\,P(g(w)), \qquad D_G(w) = 625\,P(w)^2 - 12\big(P(g(w)) + P(h(w))\big). \quad (2)$$

Lemma 2.1. *For every $w \in \mathbb{R}^5$, $C(w)C(w)^{\mathsf{T}} = C(w)^{\mathsf{T}}C(w) = C(g(w))$ and $C(w)^2 = C(h(w))$.*

Proof. Write $C = C(w)$. Substituting $m = k - j$, respectively $m = i - k$ and $m = k - i$,

$$(CC^{\mathsf{T}})_{ij} = \sum_k w_{k-i}w_{k-j} = \sum_m w_m w_{m+j-i} = g_{j-i},$$
$$(C^{\mathsf{T}}C)_{ij} = \sum_k w_{i-k}w_{j-k} = \sum_m w_m w_{m+j-i} = g_{j-i},$$

and $(C^2)_{ij} = \sum_k w_{k-i}w_{j-k} = \sum_m w_m w_{(j-i)-m} = h_{j-i}$. □

Lemma 2.2. *For every $w = (w_0, \dots, w_4) \in \mathbb{R}^5$, writing $w = (a, b, c, d, e)$,*

$$\begin{aligned} P(w) = {} & a^5 + b^5 + c^5 + d^5 + e^5 + 15\,abcde \\ & + 5\big(a^3be + a^3cd + a^2b^2d + a^2bc^2 + a^2ce^2 + a^2d^2e + ab^3c + ab^2e^2 + abd^3 + ac^3e \\ & \quad + ac^2d^2 + ade^3 + b^3de + b^2c^2e + b^2cd^2 + bc^3d + bce^3 + bd^2e^2 + c^2de^2 + cd^3e\big). \end{aligned}$$

Proof. By definition $P(w) = \sum_{\sigma \in S_5} \prod_i w_{\sigma(i)-i}$, a sum of 120 monomials of degree five; collecting them gives the 26 terms displayed. For instance, the five translations $\sigma(i) = i + k$ give w_k^5, and the monomial $abcde$ arises from the 15 permutations σ for which $i \mapsto \sigma(i) - i$ is a bijection of $\mathbb{Z}/5\mathbb{Z}$. The coefficients add up to $120 = 5!$. □

The following comparison is used to pass from the stronger inequality to Gyires's inequality.

Lemma 2.3. *For real $n \times n$ matrices M and N, $2\,\mathrm{per}(MN^{\mathsf{T}}) \le \mathrm{per}(MM^{\mathsf{T}}) + \mathrm{per}(NN^{\mathsf{T}})$.*

Proof. Let $x_1, \dots, x_n$ be indeterminates and define coefficients μ_α and ν_α, indexed by $\alpha \in \mathbb{N}^n$ with $\alpha_1 + \cdots + \alpha_n = n$, by

$$\prod_{i=1}^n \Big(\sum_{k=1}^n M_{ik}x_k\Big) = \sum_\alpha \mu_\alpha x^\alpha, \qquad \prod_{i=1}^n \Big(\sum_{k=1}^n N_{ik}x_k\Big) = \sum_\alpha \nu_\alpha x^\alpha.$$

Thus $\mu_\alpha = \sum_\varphi \prod_i M_{i\varphi(i)}$, the sum over the maps $\varphi\colon \{1, \dots, n\} \to \{1, \dots, n\}$ with occupancy α, that is, with $|\varphi^{-1}(k)| = \alpha_k$ for every k; similarly for ν_α. Expanding,

$$\mathrm{per}(MN^{\mathsf{T}}) = \sum_{\sigma \in S_n} \prod_i \sum_k M_{ik}N_{\sigma(i)k} = \sum_{\sigma \in S_n} \sum_\varphi \prod_i M_{i\varphi(i)} \prod_j N_{j,\varphi(\sigma^{-1}(j))}.$$

For fixed φ with occupancy α, the map $\sigma \mapsto \varphi \circ \sigma^{-1}$ takes every map of occupancy α exactly $\alpha! = \prod_k \alpha_k!$ times, and no other map. Hence

$$\mathrm{per}(MN^{\mathsf{T}}) = \sum_\alpha \alpha!\,\mu_\alpha\nu_\alpha. \quad (3)$$

Applying (3) also to (M, M) and (N, N), the claim follows from $2\mu_\alpha\nu_\alpha \le \mu_\alpha^2 + \nu_\alpha^2$ and $\alpha! > 0$. □

Corollary 2.4. *For every $w \in \mathbb{R}^5$, $P(h(w)) \le P(g(w))$. Consequently $D_G(w) \ge D_s(w)$.*

Proof. Take $M = C(w)$ and $N = C(w)^{\mathsf{T}}$ in Lemma 2.3. By Lemma 2.1, $MN^{\mathsf{T}} = C(h(w))$ and $MM^{\mathsf{T}} = NN^{\mathsf{T}} = C(g(w))$, so $2P(h(w)) \le 2P(g(w))$. Finally $D_G - D_s = 12\big(P(g) - P(h)\big)$. □

3 Results

Theorem 3.1. *Let $w \in \mathbb{R}^5$ with $w_i \geq 0$ for all i, and let $C = C(w)$. Then*

$$24\,\mathrm{per}(CC^{\mathsf{T}}) \leq 625\,\mathrm{per}(C)^2, \qquad \textit{that is,} \qquad \frac{5!}{5^5}\,\mathrm{per}(CC^{\mathsf{T}}) \leq \mathrm{per}(C)^2.$$

Equality holds if and only if $w_0 = w_1 = w_2 = w_3 = w_4$.

Theorem 3.2. *Let $w \in \mathbb{R}^5$ with $w_i \geq 0$ for all i, let $C = C(w)$, let π be a permutation of $\mathbb{Z}/5\mathbb{Z}$ with $w_{\pi(0)} \leq w_{\pi(1)} \leq \cdots \leq w_{\pi(4)}$, and put $z_j = w_{\pi(j)} - w_{\pi(j-1)}$ for $1 \leq j \leq 4$. Then*

$$D_s(w) = 625\,\mathrm{per}(C)^2 - 24\,\mathrm{per}(CC^{\mathsf{T}}) \;\geq\; 484\,(z_1^{10} + z_2^{10} + z_3^{10} + z_4^{10}).$$

Equality holds for $w = (1, 1, 0, 0, 0)$, for which $z_1^{10} + \cdots + z_4^{10} = 1$ and $D_s(w) = 484$. Consequently, if a real constant K satisfies $D_s(w) \geq K\,(z_1^{10} + \cdots + z_4^{10})$ for all such w and π, then $K \leq 484$.

Corollary 3.3. *Let $w \in \mathbb{R}^5$ with $w_i \geq 0$ for all i, and let $C = C(w)$. Then*

$$24\big(\mathrm{per}(CC^{\mathsf{T}}) + \mathrm{per}(C^{\mathsf{T}}C) + 2\,\mathrm{per}(C^2)\big) \leq 2500\,\mathrm{per}(C)^2,$$

with equality if and only if all five weights are equal. If moreover $w_0 + \cdots + w_4 = 1$, then $C \in \Omega_5$, the denominator below is positive, and

$$\frac{4\,\mathrm{per}(C)^2}{\mathrm{per}(CC^{\mathsf{T}}) + \mathrm{per}(C^{\mathsf{T}}C) + 2\,\mathrm{per}(C^2)} \;\geq\; \frac{24}{625} = \frac{5!}{5^5},$$

with equality if and only if $C = J_5$. Thus Conjecture 1.1 holds for every doubly stochastic circulant matrix of order five.

4 Proofs

The idea in brief. By Corollary 2.4 it suffices to bound D_s. The defect is invariant under the affine changes of indices $i \mapsto ai + b$ of $\mathbb{Z}/5\mathbb{Z}$, and these reduce the 120 possible orderings of the weights to six. In each of the six chambers, the defect written in the smallest weight and the gaps between consecutive sorted weights is a polynomial with nonnegative integer coefficients, which proves the inequality, the margin and the equality case at once.

4.1 Affine symmetry

For $a \in (\mathbb{Z}/5\mathbb{Z})^\times$ and $b \in \mathbb{Z}/5\mathbb{Z}$ let $(T_{a,b}w)_i = w_{ai+b}$; the twenty maps $i \mapsto ai + b$ are the affine bijections of $\mathbb{Z}/5\mathbb{Z}$.

Lemma 4.1. *For all $w \in \mathbb{R}^5$, a and b: $P(T_{a,b}w) = P(w)$, $g(T_{a,b}w)_k = g(w)_{ak}$ and $h(T_{a,b}w)_k = h(w)_{ak+2b}$. Consequently $D_s(T_{a,b}w) = D_s(w)$ and $D_G(T_{a,b}w) = D_G(w)$.*

Proof. We have $C(T_{a,b}w)_{ij} = w_{a(j-i)+b} = C(w)_{ai,\,aj+b}$, so $C(T_{a,b}w)$ arises from $C(w)$ by permuting the rows by $i \mapsto ai$ and the columns by $j \mapsto aj + b$; permanents are invariant under such permutations. Substituting $m = ai + b$,

$$g(T_{a,b}w)_k = \sum_i w_{ai+b}w_{a(i+k)+b} = \sum_m w_m w_{m+ak},$$
$$h(T_{a,b}w)_k = \sum_i w_{ai+b}w_{a(k-i)+b} = \sum_m w_m w_{ak+2b-m}.$$

Hence $g(T_{a,b}w) = T_{a,0}\,g(w)$ and $h(T_{a,b}w) = T_{a,2b}\,h(w)$, and the first part gives $P(g(T_{a,b}w)) = P(g(w))$ and $P(h(T_{a,b}w)) = P(h(w))$. □

4.2 Six chambers

A *canonical ordering* is one of the six sequences

$$01234, \quad 01243, \quad 01324, \quad 01342, \quad 01423, \quad 01432, \tag{4}$$

that is, an ordering $\kappa = (\kappa(0), \dots, \kappa(4))$ of $\mathbb{Z}/5\mathbb{Z}$ with $\kappa(0) = 0$ and $\kappa(1) = 1$. For a canonical ordering κ and $z = (z_0, \dots, z_4) \in \mathbb{R}^5$ let $L_\kappa(z) \in \mathbb{R}^5$ be the vector with

$$L_\kappa(z)_{\kappa(j)} = z_0 + z_1 + \cdots + z_j \qquad (0 \le j \le 4).$$

If $z \ge 0$, the weights of $L_\kappa(z)$ are nondecreasing along κ, z_0 is the smallest weight and $z_1, \dots, z_4$ are the gaps between consecutive sorted weights. The *chamber* of κ is the set $\{L_\kappa(z) : z \ge 0\}$ of nonnegative weight vectors that are nondecreasing along κ.

Lemma 4.2. *Let π be a permutation of $\mathbb{Z}/5\mathbb{Z}$. There are $a \in (\mathbb{Z}/5\mathbb{Z})^\times$, $b \in \mathbb{Z}/5\mathbb{Z}$ and a canonical ordering κ with $\pi(j) = a\kappa(j) + b$ for every j, and the triple (a, b, κ) is unique. If $w \in \mathbb{R}^5$ satisfies $w_{\pi(0)} \le \cdots \le w_{\pi(4)}$ and $w_{\pi(0)} \ge 0$, then $T_{a,b}w = L_\kappa(z)$ with $z_0 = w_{\pi(0)}$ and $z_j = w_{\pi(j)} - w_{\pi(j-1)}$ $(1 \le j \le 4)$, and $z \ge 0$.*

Proof. Put $b = \pi(0)$ and $a = \pi(1) - \pi(0) \ne 0$, and $\kappa(j) = a^{-1}(\pi(j) - b)$. Then κ is an ordering of $\mathbb{Z}/5\mathbb{Z}$ with $\kappa(0) = 0$ and $\kappa(1) = 1$, hence canonical, and $\pi(j) = a\kappa(j) + b$; conversely the conditions $\kappa(0) = 0$ and $\kappa(1) = 1$ force $b = \pi(0)$ and $a = \pi(1) - \pi(0)$. For the second statement, $(T_{a,b}w)_{\kappa(j)} = w_{a\kappa(j)+b} = w_{\pi(j)} = z_0 + \cdots + z_j$, and $z \ge 0$ because the sorted weights are nondecreasing and the smallest is nonnegative. □

4.3 The chamber certificate

Lemma 4.3. *For each canonical ordering κ, the polynomial*

$$R_\kappa(z) = D_s(L_\kappa(z)) - 484\,(z_1^{10} + z_2^{10} + z_3^{10} + z_4^{10})$$

in $z = (z_0, \dots, z_4)$ has nonnegative integer coefficients. More precisely, $D_s(L_\kappa(z))$ is homogeneous of degree ten, so it has $\binom{14}{4} = 1001$ coefficients; 996 of them are positive, and the five coefficients of z_0^{10} and $z_0^9 z_j$ $(1 \le j \le 4)$ are zero. The coefficients of $z_1^{10}, z_2^{10}, z_3^{10}, z_4^{10}$ are 233 296, 44 209, 484 and 601, and every positive coefficient is at least 484.

Proof. This is a finite computation with integer polynomials. By Lemma 2.2, $P(w)$ is an explicit integer polynomial, and by definition so is $P(g(w))$; one substitutes $w = L_\kappa(z)$ into $625P(w)^2 - 24P(g(w))$, expands, and reads off the 1001 coefficients, for each of the six orderings (4). All coefficient claims of the lemma are obtained in this way. The statement about R_κ follows: the four pure powers z_j^{10}, $j \ge 1$, have coefficients at least 484, and every other coefficient is unchanged. □

The computation can be organized so that it is small enough to be checked by a proof assistant, and this is how it is done in the formal proof (Section 5). Let $u = L_\kappa(0, z_1, \dots, z_4)$, so that $L_\kappa(z) = z_0\mathbf{1} + u$, where $\mathbf{1} = (1, \dots, 1)$. Two identities make the dependence on z_0 explicit. First, $P(t\mathbf{1} + u) = \sum_{k=0}^{5} p_k(u)\, t^k$ for explicit polynomials p_k. Second, $g(t\mathbf{1} + u) = (5t^2 + 2tS)\mathbf{1} + g(u)$, where $S = u_0 + \cdots + u_4$, so that $P(g(t\mathbf{1} + u)) = \sum_{k=0}^{5} p_k(g(u))\,(5t^2 + 2tS)^k$. Hence $D_s(L_\kappa(z)) = \sum_{j=0}^{10} Q_j\, z_0^j$ with explicit polynomials Q_j in $z_1, \dots, z_4$, and each Q_j is expanded in powers of z_1 with coefficients that are polynomials in z_2, z_3, z_4; by Lemma 4.3, $Q_9 = Q_{10} = 0$. The nonnegativity of the coefficients of R_κ amounts to the statement that, after $484\,(z_1^{10} + \cdots + z_4^{10})$ is subtracted from Q_0, each of these polynomials in z_2, z_3, z_4 has nonnegative coefficients.

4.4 Proof of Theorems 3.2 and 3.1

Proof of Theorem 3.2. Let $w \geq 0$ and π be as in the theorem, and choose a, b, κ and z by Lemma 4.2. By Lemmas 4.1 and 4.3, since $z \geq 0$,

$$D_s(w) = D_s(T_{a,b}w) = D_s\big(L_\kappa(z)\big) = 484\,\big(z_1^{10} + \cdots + z_4^{10}\big) + R_\kappa(z) \;\geq\; 484\,\big(z_1^{10} + \cdots + z_4^{10}\big).$$

For the equality case, let $w = (1, 1, 0, 0, 0)$, sorted by $\pi = (2, 3, 4, 0, 1)$, with gaps $(z_1, z_2, z_3, z_4) = (0, 0, 1, 0)$. Here $g(w) = (2, 1, 0, 0, 1)$, and Lemma 2.2 gives $P(w) = 2$ and $P(g(w)) = 84$, so $D_s(w) = 625{\cdot}4 - 24{\cdot}84 = 484$. A constant K as in the theorem therefore satisfies $K{\cdot}1 \leq 484$. □

Proof of Theorem 3.1. By Lemma 2.1, $\operatorname{per}(CC^{\mathsf{T}}) = P(g(w))$, so the inequality is $D_s(w) \geq 0$, which follows from Theorem 3.2 applied with any permutation π that sorts w. If $D_s(w) = 0$, Theorem 3.2 gives $z_1 = \cdots = z_4 = 0$, so all weights are equal. Conversely, if every weight equals c, then $P(w) = 120c^5$ (each of the 120 permutations contributes c^5) and $g(w) = (5c^2, \ldots, 5c^2)$, so $P(g(w)) = 120 \cdot 5^5 c^{10}$ and $625P(w)^2 = 9\,000\,000\,c^{10} = 24P(g(w))$. □

4.5 Proof of Corollary 3.3

Proof. By Lemma 2.1, $\operatorname{per}(CC^{\mathsf{T}}) + \operatorname{per}(C^{\mathsf{T}}C) + 2\operatorname{per}(C^2) = 2P(g(w)) + 2P(h(w))$, so the first inequality reads $4D_G(w) \geq 0$. By Corollary 2.4 and Theorem 3.1, $D_G(w) \geq D_s(w) \geq 0$. If $D_G(w) = 0$, then $D_s(w) = 0$, and all weights are equal by Theorem 3.1; conversely, for constant w one has $h(w) = g(w)$, hence $D_G(w) = D_s(w) = 0$.

Now let also $\sum_i w_i = 1$. Every row and every column of C contains each weight once, so $C \in \Omega_5$. Some weight is positive, so $g(w)_0 = \sum_i w_i^2 > 0$. The matrix $C(g(w))$ has nonnegative entries and diagonal entries $g(w)_0$, so $P(g(w)) \geq g(w)_0^5 > 0$, the identity permutation contributing $g(w)_0^5$ and the others nonnegative terms. Since also $P(h(w)) \geq 0$, the denominator $2P(g(w)) + 2P(h(w))$ is positive, and dividing the first inequality by 625 times the denominator gives the ratio inequality. Equality holds if and only if all weights are equal, that is, equal to $1/5$, which means $C = J_5$. □

5 Checking the results

A mathematical proof can contain mistakes, and this one rests on a computation too large to be checked by hand, so the results were checked in two independent ways.

Exact computations. A script for the PARI/GP computer algebra system performs 72 checks in exact integer and rational arithmetic. It was written separately from the Lean files and does not use them, or the program that generated the certificate files; it is the file `independent_checks.gp` in the directory `verification` of the accompanying repository [12]. It computes every permanent from the definition, as a sum over all permutations, and compares it with PARI's implementation of Ryser's formula. It verifies Lemmas 2.1, 2.2 and 4.1 symbolically, and that each of the 120 orderings is the image of exactly one canonical ordering under exactly one affine map (Lemma 4.2). It recomputes the chamber polynomials of Lemma 4.3, not only for the six canonical orderings but for all 120 orderings, which gives $120 \cdot 1001 = 120\,120$ coefficients, of which 119 520 are positive and 600 zero, with the zero coefficients, the least positive coefficient and the four pure-power coefficients as stated. It tests Lemma 2.3 on 200 random rational matrices of orders one to five and identity (3) on 24 random integer pairs; it tests Theorems 3.1 and 3.2 and Corollary 3.3, with their equality cases, on 2000 nonnegative weight vectors, many of them with zero or repeated weights; and it

checks the example of Theorem 3.2, the case $n = 2$ of the introduction and the remarks of Section 6. The random tests are evidence, not proofs; the chamber computation, by contrast, is exact and exhaustive.

Formal proof in Lean. Lean [8] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. The results of Section 3 are proved in Lean 4 (version 4.35.0) on top of Mathlib [10], Lean's library of formalized mathematics. The library is in the directory `lean/Circulant5` of the repository [12], in the namespace `Circulant5`. There `circ w` is the circulant with entries `w (j - i)`, `P5` the polynomial of Lemma 2.2, and `gram`, `conv` the vectors g and h; `Fin 5`, with its arithmetic modulo five, plays the role of $\mathbb{Z}/5\mathbb{Z}$.

The principal statements are restated, using only Mathlib's definitions, in a *Challenge* file, which states the theorems with their proofs omitted, the file `lean/Challenges/GyiresCirculant5.lean` (namespace `GyiresCirculant5`). There a circulant is a matrix of the form `Matrix.circulant v`, whose entries are `v (i - j)`; this is the transpose of $C(v)$, which describes the same set of matrices. Doubly stochastic matrices are Mathlib's `doublyStochastic`, the permanent is Mathlib's `Matrix.permanent`, $5!/5^5$ is the real quotient `(Nat.factorial 5 : ℝ) / 5 ^ 5`, and J_5 is the constant matrix with entries 5^{-1}. The *Solution* file `lean/Solutions/GyiresCirculant5.lean` proves the same theorems from the library. Lean's Comparator tool confirms that the theorems proved there have exactly the statements of the Challenge file and use only the standard axioms, and the proofs are accepted by Lean's kernel and by two independent kernels, NanoDa and con-ron. The table lists where each statement of the paper is proved; names in the first group are in the namespace `GyiresCirculant5`, the others in `Circulant5`.

Statement in this paper	Lean declaration(s)
Theorem 3.1	`strong_inequality`, `strong_inequality_eq_iff`
Theorem 3.2	`strong_margin`, `strong_margin_attained`, `strong_margin_sharp`
Corollary 3.3, homogeneous form	`gyires_homogeneous`, `gyires_homogeneous_eq_iff`
Corollary 3.3, ratio form	`gyires_ratio`, `gyires_ratio_eq_iff`, `gyires_denominator_pos`
Lemma 2.1	`circ_mul_transpose`, `transpose_mul_circ`, `circ_mul_self`
Lemma 2.2	`permanent_circ`
Lemma 2.3, Corollary 2.4	`PermanentGram.permanent_gram_add_ge`, `conv_permanent_le_gram`
Lemma 4.1	`P5_affine`, `gram_affine`, `conv_affine`, `strongDefect_affine`, `gyiresDefect_affine`
Lemma 4.2	`order_affine_cover`, `order_affine_cover_unique`, `canonical_chamber_of_monotone`, `exists_canonical_chamber`
Lemma 4.3, nonnegativity of R_κ	`strong_chamber_0`, ..., `strong_chamber_5`, `strong_chambers`
Theorem 3.2, the example	`strongDefect_twoSupport`, `strongDefect_margin_attained`, `strongDefect_margin_sharp`
Corollary 3.3, positive denominator	`positiveDenominator`
Section 6, the two-point example	`P5_twoSupport`, `P5_gram_twoSupport`, `P5_conv_twoSupport`, `gyiresDefect_twoSupport`, `gyiresDenominator_twoSupport`, `permanent_twoSupport_half`, `permanent_gram_twoSupport_half`, `permanent_square_twoSupport_half`, `normalizedGyiresRatio_twoSupport_half`

The Lean project was rebuilt from source, and the axioms on which each of these theorems depends were listed with Lean's `#print axioms` command. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved, and the proofs use neither `native_decide` nor any other evaluation by compiled code. The formal proof of Lemma 4.3 follows the organization described after it: the six chamber polynomials are expanded in z_0 and z_1, and each of the resulting polynomials in z_2, z_3, z_4 is written with explicit nonnegative integer coefficients, proved equal to the corresponding expansion coefficient by the `ring` tactic and nonnegative by the `positivity` tactic. These explicit coefficients were produced by a computer program, but Lean does not rely on that program: it checks every identity and every inequality itself. The formal proof of Lemma 2.3 is not the occupancy argument given here but a sum-of-squares argument over permutations, adapted from the formalization of Ozols [11].

Several remarks are also formalized: the case $n = 2$ of the introduction (`gyires_two`); the characterization of the doubly stochastic circulants of order five and their description as the combinations $\sum_k w_k Q^k$ (`circulant_mem_doublyStochastic_iff`, `circulant_eq_sum_pow`); and, for real square matrices of every order, the reduction of Gyires's inequality to the strong form described in Section 6 (`gyires_of_strong`).

What is not formalized. The formal proof establishes $R_\kappa(z) \geq 0$ for all $z \geq 0$, which is the part of Lemma 4.3 that the theorems need. The finer statements of that lemma (the numbers 996 and 5 of positive and zero coefficients, the positions of the zeros, the four pure-power

coefficients, and the least positive coefficient 484) are not stated in Lean; they are verified by the exact computation described above. The remaining remarks of Section 6, which rest on random tests or on the literature, are not formalized.

6 Discussion

Scope. The results concern circulant matrices of order five only. They do not prove Gyires's conjecture for all of Ω_5, or for any order $n \geq 6$. Within the circulants of order five they cover every matrix in the four-dimensional simplex of doubly stochastic circulants, including all of its faces, and every nonnegative circulant without normalization.

The coordinates. Written in the original weights, D_s has negative coefficients, so its nonnegativity is not visible without the change to sorted gaps. In the chamber coordinates all coefficients are nonnegative, and the zero coefficients are exactly those of the monomials z_0^{10} and $z_0^9 z_j$, which reflects the fact that D_s vanishes to second order along the constant vectors.

The two-point example. For $w = (t, t, 0, 0, 0)$ one has $g(w) = (2t^2, t^2, 0, 0, t^2)$, $h(w) = (t^2, 2t^2, t^2, 0, 0)$, $P(w) = 2t^5$, $P(g(w)) = P(h(w)) = 84t^{10}$, and $D_s(w) = D_G(w) = 484t^{10}$; this is the example showing that 484 cannot be improved. For $t = \frac{1}{2}$ the matrix is doubly stochastic, $\operatorname{per} C = \frac{1}{16}$ and $\operatorname{per}(CC^{\mathsf{T}}) = \operatorname{per}(C^2) = \frac{21}{256}$, and Gyires's ratio equals $\frac{1}{21} \approx 0.048$, above $\frac{24}{625} = 0.0384$. The example also shows that the inequality $\operatorname{per}(AA^{\mathsf{T}}) \leq \operatorname{per}(A)$ fails for some doubly stochastic circulants. If it held, the strong form $\frac{n!}{n^n} \operatorname{per}(AA^{\mathsf{T}}) \leq \operatorname{per}(A)^2$ would follow at once from van der Waerden's bound $\operatorname{per}(A) \geq n!/n^n$. According to Ryser's account, in the translation of Best and Wanless [1], an inequality of this form was once conjectured and was disproved by M. Newman with the order-four analogue $\frac{1}{2}(I + Q)$ of this example, for which $\operatorname{per} A = \frac{1}{8} < \frac{9}{64} = \operatorname{per}(AA^{\mathsf{T}})$.

A stronger question. For every real square matrix A, Lemma 2.3 with $M = A$ and $N = A^{\mathsf{T}}$ gives $2\operatorname{per}(A^2) \leq \operatorname{per}(AA^{\mathsf{T}}) + \operatorname{per}(A^{\mathsf{T}}A)$. Hence the strong form, applied to A and to A^{T}, implies Gyires's inequality for A; this is the reduction used by Ozols [11] for $n = 4$. It is therefore natural to ask whether

$$\frac{n!}{n^n} \operatorname{per}(AA^{\mathsf{T}}) \leq \operatorname{per}(A)^2 \qquad \text{for every } A \in \Omega_n.$$

This holds for $n = 2$ by the computation in the introduction, for $n = 4$ by Ozols's theorem, and for the circulants of order five by Theorem 3.1. In random tests on 2500 doubly stochastic matrices of orders $2 \leq n \leq 6$, each a convex combination of random permutation matrices, no counterexample was found.

Other orders. Every real circulant of order n is normal, and Lemma 2.3 gives $\operatorname{per}(C^2) \leq \operatorname{per}(CC^{\mathsf{T}})$ in every order, so Gyires's conjecture for circulants of order n follows from the strong form $\frac{n!}{n^n} \operatorname{per}(CC^{\mathsf{T}}) \leq \operatorname{per}(C)^2$. Random tests on 1080 nonnegative circulants of orders $2 \leq n \leq 8$ found no counterexample to this strong form. The method of this paper depends essentially on the value of n: the affine group of $\mathbb{Z}/n\mathbb{Z}$ reduces the $n!$ orderings of the weights to $(n-2)!$ chambers when n is prime, and the degree of the defect grows with n.

About this paper

How the work was done. This paper is the result of an AI research effort directed by Dev D. Goyal; the repository [12] documents the work. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It proposed the problem, found the theorems and their proofs, including the chamber certificate, wrote the Lean 4 formalization of the library `Circulant5` and the programs that generated its certificate files, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently. It ported the formalization from Lean 4.19 to Lean 4.35, adapting it to Lean's module system, rebuilt it from source, and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms. It wrote the Challenge and Solution files and the Lean files `Circulant5/MathlibStatements.lean`, which proves the margin of Theorem 3.2 for an arbitrary sorting permutation, its attainment and sharpness in that form, and the uniqueness in Lemma 4.2 (the earlier formalization stated the margin only in the six canonical chambers), and `Circulant5/Remarks.lean`, which formalizes the remarks listed in Section 5. It also wrote the PARI/GP checks and read every step of the proofs. Its checks are independent of ChatGPT's work and of the program that generated the certificate files, but not of its own Lean files and of this paper, which it wrote. It then wrote the present version of the paper, including the proofs of Lemmas 2.1, 4.1 and 4.2 in the form given here, the computation for $n = 2$ in the introduction, and the Discussion section. The theorems of Section 3 and the proof strategy are ChatGPT's.

Dev D. Goyal has reviewed the proofs and is responsible for this paper; this review is not an independent expert validation.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal.

Priority. The searches for earlier work covered the works cited here, as far as they could be accessed, general web searches for Gyires's conjecture and for permanent inequalities for circulant matrices, the account of prior work in Ozols's repository [11], and the Lean collections google-deepmind/formal-conjectures (commit `37f0808`, 24 October 2026), openai/math (commit `1581d4c`, 12 October 2026) and plby/lean-proofs (commit `8822f7d`, 15 September 2026); none of them contains the circulant case. Gyires's original article, the full texts of the survey of Cheon and Wanless and of Chang's article, and Lavi's catalogue could not be accessed; the statement of Conjecture 1.1 and the results reported from them were checked against the account of prior work in Ozols's repository [11], against the publishers' abstracts, and against other citations of these works. These searches were limited, and no claim of priority or novelty is made.

Corrections. Please report any error in the issue tracker of the repository [12], where the Lean sources and the numerical checks are also available.

Acknowledgements

The conjecture is due to B. Gyires. The formal proof of Lemma 2.3 is adapted from the Lean formalization of the case $n = 4$ by C. Ozols [11], which is licensed under the Apache License 2.0. Thanks to the developers of Lean and Mathlib.

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